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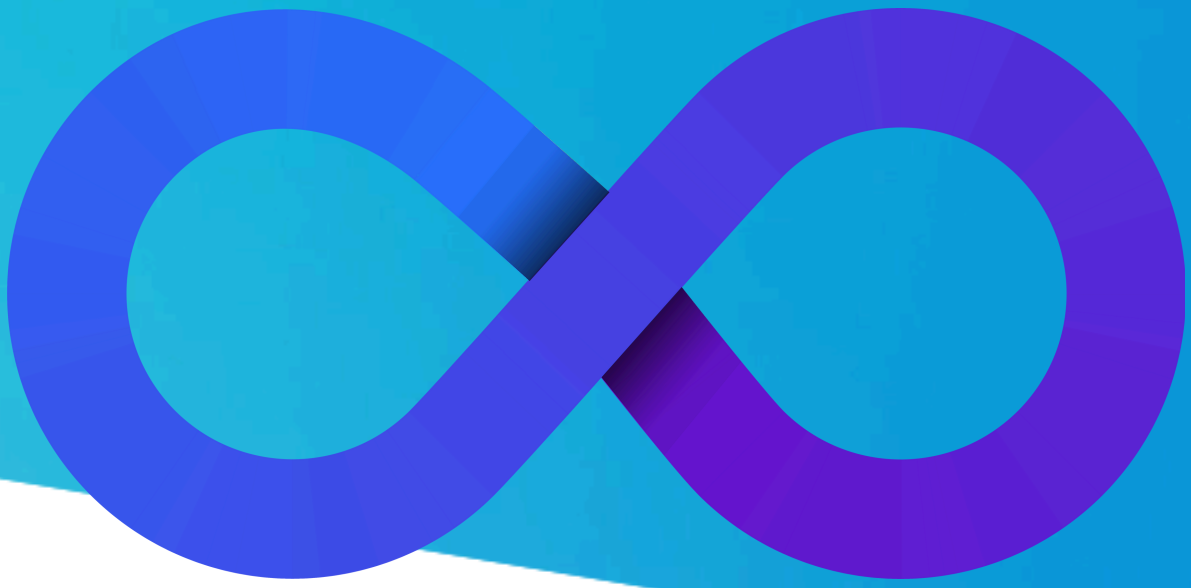
DevOps with AI

Advanced Certification Program

Curriculum



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Module 1: Core Concepts (DevOps Certification)

- Devops Fundamentals
- GitOps: Git, GitHub, GitLab
- Infrastructure as Code (IaC): Terraform, AWS CloudFormation
- Microservices: Kubernetes, Docker
- DevSecOps: Jenkins (with security plugins), GitLab CI/CD (with security pipelines)

Module 2: CI/CD Tools

- Jenkins: Job creation, pipeline as code, distributed builds (master-slave architecture)
- GitLab CI/CD: CI/CD pipelines, multi-project pipelines
- GitHub Actions: Advanced workflows, automation tasks

Module 3: Containerisation with Docker & Kubernetes (Docker & Kubernetes Certification)

- Docker: DockerHub, Docker networking, Docker Swarm
- Kubernetes: Pods, Services, Deployments, StatefulSets, ConfigMaps, Secrets
- Istio: Service mesh for Kubernetes, traffic management, security with mTLS
- Tekton: Building CI/CD pipelines in Kubernetes
- FluxCD and Argo CD: GitOps tools for continuous delivery

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Module 4: DevOps Cloud Certification

- Cloud-Native Development: Twelve-Factor App methodology, serverless architectures
- AWS Services: Various computing, storage, networking, and database services
- Azure Services: Virtual machines, Azure Kubernetes Service (AKS), Azure Functions
- Continuous Delivery: Canary releases, blue/green deployments
- Security: Zero trust model, mTLS for service-to-service communication

Module 5: Infrastructure as Code (IaC) (Terraform Certification)

- Ansible : Basic and Advanced
- Terraform: Providers, resources, variables, output blocks, dynamic blocks, modules, workspaces, remote state management
- Terraform Cloud: Collaboration, scalability, governance features
- AWS and Azure: Integration with Terraform, infrastructure provisioning

Module 6: Monitoring, Logging, and Observability

- Monitoring: Prometheus, Grafana, Nagios XI
- Logging: ELK Stack (Elasticsearch, Logstash, Kibana)

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- Advanced Monitoring: Prometheus (advanced features, alerting), Grafana (advanced features), Jaeger (distributed tracing)
- Alerting and Dashboards: Prometheus Alertmanager, Grafana dashboards
- Scaling Monitoring Infrastructure: Strategies for Prometheus and Grafana

Module 7: DevOps with AI (DevOps with AI Certification)

- AI-Driven Automation
- Predictive Analytics for DevOps
- Incident Management and Root Cause Analysis
- Enhancing Security with AI in DevOp(AIOps)
- AI-Enhanced Monitoring and Observability
- AI-Driven Optimization and Self-Healing Systems

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Additional Tools and Technologies Covered

- Version Control: Git, GitHub, GitLab
- CI/CD: Jenkins, GitLab CI/CD, GitHub Actions
- Containerization: Docker
- Container Orchestration: Kubernetes
- Service Mesh: Istio
- Infrastructure as Code: Terraform
- Cloud Platforms: AWS, Azure
- Monitoring and Observability: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Jaeger
- GitOps Tools: FluxCD, Argo CD
- Serverless Computing: AWS Lambda, Azure Functions
- Security Tools: Various security plugins integrated into CI/CD pipelines

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1 DevOps Core Concepts

Basic:

- **Session 1: Introduction to DevOps**
    - Introduction Introduction to Software Development and DevOps
    - DevOps Lifecycle and Tools
    - Introduction to DevOps on Cloud
    - Linux Fundamentals for DevOps

- **Session 2: Introduction to Advanced DevOps Practices**
    - Evolution of DevOps: From CI/CD to GitOps
        - Historical timeline and key milestones in the evolution of DevOps.
        - Evolution from CI/CD practices to the emergence of GitOps.
        - Impact of DevOps on software development lifecycle and operations.
    - Historical Overview of DevOps
    - Benefits and challenges of GitOps
        - Definition and core principles of GitOps (declarative infrastructure, version control as the single source of truth).
        - Benefits of GitOps: efficiency, traceability, repeatability, and scalability.
        - Challenges of GitOps adoption: cultural shift, tooling complexity, and security considerations.

Case Studies:

- Examples of organizations (e.g., Netflix, Spotify) leveraging GitOps for operational excellence.

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1 DevOps Core Concepts

- **Session 3: Understanding Infrastructure as Code (IaC)**
    - Definition and importance of IaC
        - Principles of IaC: automation, consistency, versioning, and scalability.
        - Benefits of IaC in modern IT environments: agility, reproducibility, and cost-effectiveness.
    - Common tools and frameworks (Terraform, CloudFormation)
        - Overview and comparison of Terraform and AWS CloudFormation.
        - Practical demonstrations of using Terraform to provision infrastructure.

Intermediate:

- **Session 4: Principles of Microservices Architecture**

- Introduction to Microservices Architecture
    - Principles and characteristics: modularity, bounded contexts, and decentralized data management.
    - Scalability, resilience, and fault isolation benefits of microservices.
- How microservices influence DevOps practices
    - CI/CD pipelines for microservices: independent deployments and continuous integration challenges.
    - Service discovery, load balancing, and monitoring considerations.
- Comparing monolithic and microservice architecture
    - Contrasting characteristics: deployment complexity, scaling limitations, and technology stack flexibility.
    - Case studies of successful microservices migrations.

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1 DevOps Core Concepts

- **Session 5: Introduction to DevSecOps**
  - Principles of DevSecOps
    - Shift-left security: integrating security throughout the software development lifecycle (SDLC).
    - Security automation: vulnerability scanning, static code analysis, and continuous monitoring.
  - Comparison of DevSecOps tools with traditional DevOps tools
    - Traditional DevOps vs. DevSecOps tooling: Jenkins vs. Jenkins with security plugins, GitLab CI/CD security pipelines.
    - Container security tools (e.g., Docker Bench, Clair) and infrastructure security as code practices.
  - DevSecops
    - High-profile security breaches and lessons learned.
    - Implementing DevSecOps in regulatory environments (e.g., PCI-DSS compliance).

- **Session 6: Basic Git Workflow**
  - Add workflow, commits, revert, reset, tags, gitignore
    - Introduction to Git: version control, repositories, commits, and branches.
    - Understanding Git terminology: commits, branches, tags, and merging.
    - Revert, reset, and amend commits.
    - Ignoring files
    - Branching strategies: feature branching, release branching, and hotfixes.
    - Git collaboration workflows: GitFlow, GitHub Flow, and GitLab Flow.
    - Group exercise: collaborating on a Git repository using branching and merging strategies.
    - Resolving conflicts and handling code reviews in Git-based workflows.

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1 DevOps Core Concepts

Advanced:

- **Session 7: Advanced Git Usage**
    - Git internals and advanced branching strategies
        - Understanding Git objects: blobs, trees, commits, and tags.
        - Exploring the Git repository structure and object hashing.
    - GitFlow, GitHub Flow, GitLab Flow
        - GitFlow: feature branches, release branches, and hotfix branches.
        - GitHub Flow: continuous deployment with mainline branches.
        - GitLab Flow: leveraging CI/CD pipelines with version control.
        - Rebase vs. merge strategies: interactive rebase and squash commits.
        - Git hooks: automating pre-commit and post-commit actions.
- **Session 8: Git Workflow Patterns for Collaboration and Release Management**
    - Release Branching and Release Tagging Strategies
        - Semantic versioning: tagging releases and managing changelogs.
        - Release branching models: trunk-based development vs. feature branches.
    - Git Workflow Automation with GitHub Actions and GitLab CI/CD
        - Integrating Git with CI/CD pipelines: automating builds, tests, and deployments.
        - Leveraging GitHub Actions and GitLab CI/CD for workflow automation.
        - Declarative configuration management with GitOps principles.
        - Continuous delivery with infrastructure as code (IaC) and version-controlled environments.

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1 DevOps Core Concepts

- **Session 9: Case Studies and Practical Labs**
    - Real-world case studies
        - Hands-on lab: configuring CI/CD pipelines for Git-based workflows.
        - Hands-on lab: Applying Git, IaC, and CI/CD principles to solve practical challenges.
    - Practical exercises
        - Analyzing case studies of successful GitOps implementations in enterprise environments.

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2 CI/CD Tools

Basic:

- **Session 10: Mastering Jenkins**

- Job creation, plugin management
    - Overview of Jenkins as a CI/CD tool.
    - Installation, setup, and basic configuration.
    - Creating and configuring Jenkins jobs: freestyle vs. pipeline jobs.
    - Best practices for job naming conventions, parameterization, and job dependencies.
    - Exploring essential Jenkins plugins for various functionalities (e.g., Git, Docker, SonarQube).
    - Managing plugin updates, installation, and compatibility.
    - Configuring authentication and authorization in Jenkins.
    - Implementing security best practices for Jenkins instances.

Hands-on Lab:

- Setting up a Jenkins server.
- Creating and configuring Jenkins jobs with plugins for automated testing and deployment.

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2 CI/CD Tools

Intermediate:

- **Session 11: Jenkins Pipeline as Code**
  - Declarative pipeline syntax
    - Introduction to Jenkinsfile: structure, stages, and steps.
    - Writing declarative pipelines for building, testing, and deploying applications.
  - Integrating security tools into CI/CD pipelines
    - Integrating static code analysis tools (e.g., SonarQube, Checkmarx) into CI/CD pipelines.
    - Implementing security gates and automated vulnerability assessments.
    - Version controlling Jenkinsfile: best practices for managing pipeline code.
    - Automated testing and validation of pipeline changes.

Hands-on Lab:

- Creating a Jenkins Pipeline using declarative syntax.
- Integrating security checks into the pipeline and configuring automated tests.

Advanced:

- **Session 12: Advanced Jenkins Configurations**
  - Distributed Builds and Scalability with Jenkins Master-Slave Architecture
    - Setting up Jenkins master-slave architecture for parallel builds.
    - Load balancing and scaling Jenkins agents for distributed environments.

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2 CI/CD Tools

- Advanced Pipeline Configuration: Variables, caching, and retries
  - Using environment variables, caching dependencies, and retry mechanisms in pipelines.
  - Handling failures gracefully and implementing retry strategies.
  - Monitoring and optimizing Jenkins performance: memory allocation, disk space management.
  - Configuring Jenkins for high availability and disaster recovery.

- **Hands-on Lab:**

  - Configuring Jenkins master-slave setup.
  - Implementing advanced pipeline features like caching and retry mechanisms.

- **Session 13: Multi-project Pipelines and Cross-project Dependencies**

  - GitLab CI/CD Pipelines
    - Orchestrating pipelines across multiple repositories and projects.
    - Managing dependencies and triggering builds across interconnected projects.
    - Introduction to GitLab CI/CD: CI/CD configuration in GitLab CI/CD YAML.
    - Leveraging GitLab CI/CD for multi-project pipelines and integration with GitLab features (e.g., Merge Requests).
  - GitHub Actions: Advanced workflows
    - Advanced workflows in GitHub Actions: matrix builds, conditional steps, and environment deployments.

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2 CI/CD Tools

- Advanced workflows in GitHub Actions: matrix builds, conditional steps, and environment deployments.
- Integrating GitHub Actions with GitHub features (e.g., Pull Requests, Code Reviews).

- **Hands-on Lab:**

- Implementing multi-project pipelines using GitLab CI/CD.
- Setting up advanced workflows and automation tasks in GitHub Actions.

Practical Labs:

- **Session 14-16: Hands-on Labs**

- Practical labs for Jenkins, GitLab CI/CD, GitHub Actions
  - Rotating labs covering Jenkins basics, Pipeline as Code, advanced configurations, and multi-project pipelines.
  - Practical exercises aligning with industry-standard use cases and deployment scenarios.
  - Simulated deployment scenarios: continuous integration, automated testing, and deployment to different environments.
  - Troubleshooting common issues in CI/CD pipelines and optimizing pipeline performance.
  - Peer review sessions: sharing insights and optimizing CI/CD pipelines based on feedback.
  - Iterative development: refining pipeline scripts and configurations to meet evolving project requirements.

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3 Containerization with Docker & Kubernetes

Basic:

- **Session 17: Introduction to Docker**
  - Docker Images and DockerHub
    - Overview of containerization: benefits and use cases in software development and deployment.
    - Introduction to Docker architecture: Docker daemon, Docker client, and Docker registry (DockerHub).
  - Dockerfile and building custom images
    - Creating Docker images: Dockerfile syntax, layers, and best practices.
    - Publishing and managing Docker images on DockerHub.
  - Docker Port Mapping
    - Writing Dockerfiles for building custom Docker images.
    - Best practices for optimizing Docker images: minimizing image size and improving build efficiency.
    - Basics of container networking: Docker network drivers, bridge networks, and host networking.
    - Use cases for container communication and network isolation.

Intermediate:

- **Session 18: Docker Networking**
  - Container Networking Interfaces (CNIs)
    - Overview of CNIs and their role in container networking.
    - Implementing custom network configurations using CNIs.
    - Configuring overlay networks for multi-host communication in Docker Swarm and Kubernetes.
    - Security considerations and network policies in containerized environments.

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3 Containerization with Docker & Kubernetes

- Using Docker Swarm and Kubernetes for service discovery and load balancing.
- Deploying applications with multiple containers and managing inter-container communication.

- **Session 19: Docker Swarm Mode**
  - Introduction and basic configurations
    - Overview of Docker Swarm architecture: manager nodes, worker nodes, and service orchestration.
    - Deploying services in Docker Swarm: stack deployment, service scaling, and rolling updates.
    - Configuring overlay networks in Docker Swarm for multi-host communication.
    - Load balancing and service discovery with Swarm mode.
    - Implementing high availability strategies in Docker Swarm: manager node redundancy and fault tolerance.
    - Handling node failures and ensuring service continuity.

- **Session 20: Kubernetes Components and Architecture**

- Overview of Kubernetes architecture
  - Master and worker nodes: kube-apiserver, kube-controller-manager, kube-scheduler, etcd, and kubelet.
  - Understanding Kubernetes cluster architecture and control plane components.
- Kubernetes Pods and multi-container Pods
  - Anatomy of a Pod: containers, shared storage volumes, and networking.
  - Deploying multi-container Pods and managing inter-container communication.
  - Strategies for deploying applications with Kubernetes Deployments.
  - Rolling updates, rollback strategies, and managing application versions.

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3 Containerization with Docker & Kubernetes

- **Session 21: Kubernetes Services and Deployments**

  - Services: Types and configurations
    - Types of Kubernetes Services: ClusterIP, NodePort, LoadBalancer, and ExternalName.
    - Service discovery and load balancing within a Kubernetes cluster.
  - Deployments: Strategies and use cases
    - Using Kubernetes Deployments for declarative application management.
    - Strategies for zero-downtime deployments, blue-green deployments, and canary releases.
    - ConfigMaps and Secrets: managing application configurations and sensitive data in Kubernetes.
    - Using Helm for managing Kubernetes application packages and templating.

Advanced:

- **Session 22: Kubernetes Storage and Persistent Volume**

  - Stateful sets, Config Maps, and Secrets
    - Deploying stateful applications with Kubernetes StatefulSets.
    - Managing pod identity, persistent storage, and ordered deployment and scaling.
    - Overview of Kubernetes Persistent Volumes: storage classes, volume plugins, and dynamic provisioning.
    - Configuring and managing PVs and Persistent Volume Claims (PVCs).
    - Managing application configurations and sensitive data using ConfigMaps and Secrets.
    - Best practices for securing and accessing secrets in Kubernetes environments.

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3 Containerization with Docker & Kubernetes

- **Session 23: Advanced Kubernetes Scheduling Techniques**
    - Affinity, Anti-affinity, and Taints/Tolerations
        - Using node affinity and pod affinity/anti-affinity to influence pod placement in Kubernetes.
        - Strategies for optimizing application performance and availability using affinity rules.
- Resource Management: Resource Quotas, Limit Ranges
    - Kubernetes resource quotas: defining resource limits and usage constraints for namespaces.
    - Limit ranges: configuring default resource requests and limits for Kubernetes pods.

- **Session 24: Kubernetes in Production**
    - Advanced deployments: Canary, Blue/Green
        - Canary deployments: gradually rolling out new versions and gathering user feedback.
        - Blue-green deployments: switching traffic between multiple identical environments.
        - Autoscaling: Horizontal Pod Autoscaler, Cluster Autoscaler, Vertical Pod Autoscaler
        - Horizontal Pod Autoscaler (HPA): automatic scaling based on CPU or custom metrics.
        - Cluster Autoscaler: scaling Kubernetes clusters based on resource usage and pending pods.
        - Vertical Pod Autoscaler: optimizing resource allocation for individual pods based on resource usage patterns.
        - Implementing monitoring solutions (e.g., Prometheus, Grafana) for Kubernetes clusters.
        - Configuring logging with Elasticsearch, Fluentd, and Kibana (EFK stack) for centralized log management.

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3 Containerization with Docker & Kubernetes

- **Session 25: Kubernetes with Istio**
  - Traffic Management
    - Overview of Istio architecture: Envoy proxy, Mixer, Pilot, and Istio control plane.
    - Installing and configuring Istio in a Kubernetes cluster.
  - Request Routing, Traffic Shifting, and Load Balancing
    - Implementing request routing, traffic shifting, and load balancing with Istio VirtualServices.
    - Canary deployments and A/B testing using Istio traffic management features.
    - Securing service-to-service communication with Istio mTLS (mutual TLS) authentication.
    - Distributed tracing with Jaeger for monitoring and troubleshooting microservices interactions.
- **Session 26: Tekton and GitOps**
  - Building CI/CD Pipelines with Tekton
    - Introduction to Tekton: architecture, Tasks, Pipelines, and PipelineRuns.
    - Building and managing CI/CD pipelines with Tekton in Kubernetes clusters.
  - Implementing GitOps with FluxCD and Argo CD
    - GitOps principles and practices: declarative infrastructure and continuous delivery.
    - Setting up and configuring FluxCD for automated GitOps workflows.
    - Managing application deployments with Argo CD and GitOps best practices.

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3 Containerization with Docker & Kubernetes

Practical Labs:

- **Session 27-28: Hands-on Labs**
    - Practical labs for GitOps, Docker and Kubernetes
        - Rotating labs covering Docker basics, Kubernetes networking, storage, advanced scheduling, Istio traffic management, and GitOps principles.
        - Practical exercises aligning with industry-standard use cases and deployment scenarios.
    - Scenario-based Exercises:
        - Simulated deployment scenarios: multi-container applications, stateful applications with persistent storage, and advanced deployment strategies.
        - Troubleshooting common issues in Kubernetes deployments and optimizing application performance.

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4 Cloud Certification

Basic:

- **Session 29: Cloud-Native Development Patterns**
  - Twelve-Factor App Methodology and Cloud-Native Design Patterns
    - Principles of twelve-factor apps: best practices for building scalable, maintainable applications.
    - Understanding each factor: environment parity, dependencies management, configuration, and backing services.
    - Design patterns for microservices and serverless architectures.
    - Patterns for resilience, scalability, and observability in cloud-native applications.
    - Analyzing benefits and challenges of adopting cloud-native development approaches.

Intermediate:

- **Session 30: Building Cloud-Native Applications**
  - With Kubernetes and Serverless Architectures
    - Container orchestration with Kubernetes: deployment strategies and scaling applications.
    - Using Kubernetes for managing microservices and distributed applications.
    - ntroduction to serverless computing: benefits and limitations.
    - Implementing serverless applications on cloud platforms (e.g., AWS Lambda, Google Cloud Functions).
    - Integrating Kubernetes and serverless components in hybrid cloud environments.
    - Choosing between Kubernetes and serverless based on application requirements and use cases.

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4 Cloud Certification

- **Session 31: Implementing Continuous Delivery for Cloud-Native Applications**

  - Progressive Delivery Techniques: Canary Releases, Blue/Green Deployments
    - Principles and benefits of continuous integration and continuous deployment (CI/CD) for cloud-native apps.
    - Building automation pipelines for CI/CD in cloud environments.
    - Canary releases: gradual rollout and testing of new features in production.
    - Blue/green deployments: switching traffic between multiple environments for zero-downtime updates.Implementing feature toggles and A/B testing strategies in cloud-native applications.
    - Managing feature flags dynamically and their impact on continuous delivery pipelines.

Intermediate:

- **Session 32: Securing Cloud-Native Applications**

  - Zero Trust Security Model for Cloud-Native Environments
    - Principles of zero trust security for cloud-native environments.
    - Implementing identity-based access controls and least privilege principles.
  - Secure Service-to-Service Communication with mTLS and SPIFFE/SPIRE
    - Mutual TLS (mTLS) authentication for securing communication between microservices.
    - Using SPIFFE (Secure Production Identity Framework for Everyone) and SPIRE (SPIFFE Runtime Environment) for identity management.

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4 Cloud Certification

- Securing APIs, containers, and orchestrators in cloud environments.
- Monitoring and auditing for compliance and threat detection in cloud-native applications.

Intermediate:

- **Session 33: Overview of AWS and Azure**
  - Key services and DevOps integrations
    - Overview of core AWS services: compute, storage, networking, databases, and analytics.
    - Integration of AWS services with DevOps tools and practices.
    - Overview of key Azure services: virtual machines, Azure Kubernetes Service (AKS), Azure Functions, and Azure databases.
    - Azure DevOps: CI/CD pipelines, integration with Azure services, and infrastructure as code (IaC) with Azure Resource Manager (ARM).
    - Contrasting AWS and Azure offerings in terms of scalability, pricing, and service-specific features.
    - Choosing the right cloud platform based on application requirements and organizational needs.

Practical Labs:

- **Session 34: Hands-on Labs**
  - Practical lab for cloud-native development and security
    - Setting up cloud-native applications using Kubernetes and serverless technologies.
    - Implementing CI/CD pipelines for cloud-native applications.
    - Securing microservices communication with mTLS and exploring zero trust security models.

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5 Infrastructure as Code (IaC)

Basics:

- **Session 35: Terraform Basics**
    - Providers, Resources, Variables, Output Blocks, Dynamic Blocks
        - Overview of Infrastructure as Code (IaC) principles and benefits.
        - Understanding Terraform's declarative syntax and its advantages.
        - Terraform Configuration Language (HCL)
            - Writing Terraform configuration files: providers, resources, variables, and output blocks.
            - Managing infrastructure components (AWS, Azure, Google Cloud) using Terraform.
        - State Management:
            - Importance of Terraform state: tracking resources and managing changes.
            - State file formats, state locking, and handling concurrent Terraform operations.
        - Dynamic Blocks:
            - Utilizing dynamic blocks in Terraform configurations for flexible resource definitions.
            - Best practices for using dynamic blocks effectively.

Advanced:

- **Session 36: Terraform Modules and Workspaces**
    - Creating reusable modules
        - Terraform Modules:
            - Creating and using Terraform modules for reusable infrastructure components.
            - Structuring modules for scalability and maintainability.

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4 Infrastructure as Code (IaC)

- Managing environments with workspaces
  - Overview of Terraform Cloud: benefits for collaboration, remote state management, and version control.
  - Integrating Terraform Cloud with CI/CD pipelines and version control systems.
- Terraform Cloud: Collaboration and Remote State Management

Advanced:

- **Session 37: Advanced Terraform Features**

  - Collaboration features, Remote State Management, and Locking
    - Collaboration Features:
      - Implementing collaboration workflows with Terraform Enterprise or Terraform Cloud.
      - Role-based access control (RBAC), audit logs, and managing Terraform configurations across teams.
    - Remote State Management:
      - Strategies for remote state storage: Terraform Cloud, AWS S3, Azure Blob Storage.
      - Enhancing security and scalability with remote state management.
    - Terraform Enterprise: Scalability and Governance
      - Deploying Terraform Enterprise for large-scale infrastructure management.
      - Governance practices: policy as code, Sentinel policies, and compliance enforcement.

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5 Infrastructure as Code (IaC)

Practical Labs:

- **Session 38: Hands-on Lab**
    - Practical lab for Terraform
        - Setting up Terraform environment: installing Terraform CLI, configuring providers, and initializing projects.
        - Creating Terraform configurations for provisioning infrastructure resources (e.g., virtual machines, networks, storage).
        - Managing state files, workspaces, and modules in Terraform.
        - Integrating Terraform with CI/CD pipelines for automated infrastructure deployments.
        - Best practices for using dynamic blocks effectively.

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5 Monitoring, Logging, and Observability

Basic:

- **Session 39: Introduction to Monitoring and Logging**

  - Basic Monitoring with Prometheus,Grafana,Nagios XI
    - Basic Monitoring:
      - Overview of monitoring principles: metrics, metrics collection, and visualization.
      - Introduction to Prometheus:
        - Installation and configuration.
        - Defining and querying metrics.
      - Grafana for visualization:
        - Setting up Grafana dashboards.
        - Creating visualizations and charts.
  - Overview of Nagios XI for alerting and monitoring infrastructure health.
  - Basic Logging with ELK Stack (Elasticsearch, Logstash, Kibana)
    - Overview of ELK Stack components:
      - Elasticsearch: indexing and storing logs.
      - Logstash: log aggregation and parsing.
      - Kibana: log search and visualization.
  - Configuring Logstash pipelines for log processing.
  - Creating dashboards and visualizations in Kibana.

Advanced

- **Session 40: Advanced Monitoring and Observability**

  - Advanced Monitoring with Prometheus and Grafana
    - Prometheus advanced features:
      - Service discovery mechanisms.
      - Recording rules and alerts.
    - Advanced Grafana features:
      - Templating and variables.
      - Dashboard annotations and annotations.

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7 DevOps with AI

Basic:

- **Session 41: Introduction to AI in DevOps**
  - Overview of AI and its relevance to DevOps
  - Benefits of integrating AI in DevOps: automation, predictive analytics, and enhanced decision-making
  - Key AI techniques and tools for DevOps: machine learning, neural networks, and natural language processing
  - Case studies: How leading organizations use AI in DevOps to improve efficiency and reliability

- **Session 42: AI-Driven Automation in DevOps**
  - AI in CI/CD pipelines: automating testing, deployment, and monitoring
  - Leveraging AI for code quality analysis and bug detection
  - Tools and frameworks for AI-driven automation: TensorFlow, PyTorch, and OpenAI tools
  - Practical demonstration: Implementing AI-driven automation in a sample CI/CD pipeline

Intermediate:

- **Session 43: Predictive Analytics for DevOps**
  - Introduction to predictive analytics and its role in DevOps
  - Use cases: performance forecasting, capacity planning, and anomaly detection
  - Tools and platforms for predictive analytics in DevOps: DataRobot, Splunk, and ELK Stack with machine learning plugins
  - Practical demonstration: Implementing predictive analytics for performance monitoring.

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7 DevOps with AI

- **Session 44: AI for Incident Management and Root Cause Analysis**
    - AI-driven incident management: reducing MTTR with automated diagnostics
    - Using machine learning for root cause analysis: techniques and best practices
    - Tools for AI-driven incident management: IBM Watson AIOps, Moogsoft, and BigPanda
    - Practical demonstration: Implementing AI-driven incident management and root cause analysis

- **Session 45: Enhancing Security with AI in DevOps (AIOps)**
    - Introduction to AIOps: AI for IT operations and security
    - Using AI for threat detection, vulnerability management, and compliance
    - Tools for AIOps: Darktrace, Sumo Logic, and Splunk with AI integrations
    - Practical demonstration: Implementing AI for security monitoring and threat detection

- **Session 46: AI-Enhanced Monitoring and Observability**
    - AI for monitoring: anomaly detection, pattern recognition, and predictive alerts
    - Integrating AI with existing monitoring tools: Prometheus, Grafana, and ELK Stack
    - Case studies: Real-world examples of AI-enhanced monitoring in DevOps
    - Practical demonstration: Setting up AI-enhanced monitoring for a sample application.

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7 DevOps with AI

Advanced

- **Session 47: AI-Driven Optimization and Self-Healing Systems**
  - AI for resource optimization: dynamic scaling and resource allocation
  - Implementing self-healing systems with AI: automated remediation and fault tolerance
  - Tools for AI-driven optimization: Kubernetes with AI integrations, Google Cloud AI
  - Practical demonstration: Building a self-healing system using AI and Kubernetes.

- **Session 48: Practical Labs and Real-World Applications**
  - Hands-on lab: Implementing AI-driven CI/CD pipeline
  - Hands-on lab: AI for predictive analytics and performance monitoring
  - Hands-on lab: AI-driven incident management and root cause analysis
  - Analyzing case studies of successful AI integration in DevOps.

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